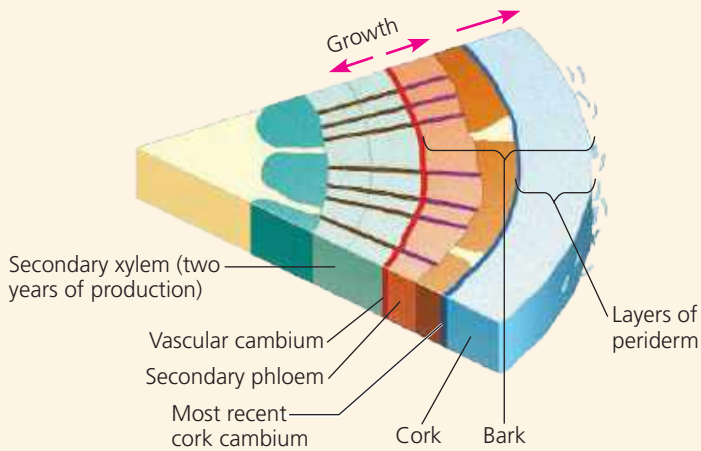


- The **cork cambium** gives rise to a thick protective covering called the periderm, which consists of the cork cambium plus the layers of cork cells it produces.



? What advantages did plants gain from the evolution of secondary growth?

CONCEPT 35.5

Growth, morphogenesis, and cell differentiation produce the plant body (pp. 775–780)

- Cell division and cell expansion are the primary determinants of growth.
- Morphogenesis, the development of body shape and organization, depends on cells responding to positional information from their neighbors.
- Cell differentiation, arising from differential gene activation, enables cells within the plant to assume different functions despite having identical genomes. The way in which a plant cell differentiates is determined largely by the cell's position in the developing plant.
- Internal or environmental cues may cause a plant to switch from one developmental stage to another—for example, from developing juvenile leaves to developing mature leaves. Such changes in shape are called **phase changes**.
- Studies of floral development have provided a model system for studying **pattern formation**. The **ABC hypothesis** identifies how three classes of genes control formation of sepals, petals, stamens, and carpels.

? By what mechanism do plant cells tend to elongate along one axis instead of expanding in all directions?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Most of the growth of a plant body is the result of
 - cell differentiation.
 - morphogenesis.
 - cell division.
 - cell elongation.
- The innermost layer of the root cortex is the
 - core.
 - pericycle.
 - endodermis.
 - pith.

- Heartwood and sapwood consist of
 - bark.
 - periderm.
 - secondary xylem.
 - secondary phloem.
- The phase change of an apical meristem from the juvenile to the mature vegetative phase is often revealed by
 - a change in the shape of the leaves produced.
 - the initiation of secondary growth.
 - the formation of lateral roots.
 - the activation of flower-inducing genes.
- The vascular cambium gives rise to
 - all xylem.
 - all phloem.
 - primary xylem and phloem.
 - secondary xylem and phloem.
- The root pericycle is the site where
 - secondary growth originates.
 - root hairs originate.
 - lateral roots originate.
 - the endodermis originates.
- Root apical meristems are found
 - only in taproots.
 - only in lateral roots.
 - only in adventitious roots.
 - in all roots.

Levels 3-4: Applying/Analyzing

- Suppose a flower had normal expression of genes A and C and expression of gene B in all four whorls. Based on the ABC hypothesis, what would be the structure of that flower, starting at the outermost whorl?
 - carpel-petal-petal-carpel
 - petal-petal-stamen-stamen
 - sepal-carpel-carpel-sepal
 - sepal-sepal-carpel-carpel
- Which of the following arise(s), directly or indirectly, from meristematic activity?
 - secondary xylem
 - leaves
 - dermal tissue
 - all of the above
- A strawberry plant mutant that fails to make stolons would suffer from
 - too little mineral absorption.
 - a tendency to topple over.
 - too little water absorption.
 - a reduction in asexual reproduction.
- DRAW IT** On this cross section from a woody eudicot, label a growth ring, late wood, early wood, and a vessel element. Then draw an arrow in the pith-to-cork direction.

